

React Roadmap: 100 Topics (Beginner to Advanced)

Basics (Beginner Level)

- What is React?
- JSX (JavaScript XML)
- Components (Functional & Class)
- Rendering Elements
- Props (Properties)
- State (useState)
- Handling Events
- Conditional Rendering
- Lists and Keys
- Forms and Inputs
- Controlled vs Uncontrolled Components
- Lifting State Up
- Basic CSS in React
- Inline Styling
- Component Reusability
- Fragment (<></>)
- Default Props
- PropTypes for type checking
- React Strict Mode
- Basic Folder Structure in React App

Hooks (Intermediate Level)

- useState
- useEffect
- useRef
- useContext
- useReducer
- useMemo
- useCallback
- Custom Hooks

- Rules of Hooks
- Hook Dependencies

Routing

- React Router DOM (v6+)
- BrowserRouter, Routes, Route
- Link & NavLink
- useNavigate
- useParams
- useLocation
- Nested Routes
- Route Redirection
- Protected Routes (Private Routes)
- 404 Pages & Wildcard Routing

State Management

- Context API
- Redux (Basic)
- Redux Toolkit (RTK)
- Redux Thunk
- Redux DevTools
- useSelector & useDispatch
- Global State vs Local State
- Zustand (Lightweight alternative)
- Recoil (State management library)
- MobX (Alternative to Redux)

API & Data Fetching

- Fetch API
- Axios
- useEffect with Fetch
- Loading and Error States
- useSWR (React Hooks for data fetching)
- React Query (TanStack Query)

- Debouncing & Throttling APIs
- Pagination & Infinite Scroll
- Handling async/await in React

Performance Optimization

- Memoization (useMemo, React.memo)
- useCallback optimization
- React Lazy & Suspense (Code Splitting)
- Virtual DOM understanding
- Avoiding unnecessary re-renders
- Optimizing lists with keys
- useTransition (React 18)
- Concurrent Mode (React 18)

Advanced Patterns

- Compound Components
- Render Props
- Higher Order Components (HOC)
- Controlled vs Uncontrolled Inputs
- Portals
- Error Boundaries
- Refs and DOM manipulation
- Forward Refs
- Context with useReducer pattern
- Dynamic Import (lazy loading components)

Testing in React

- React Testing Library
- Jest Basics
- Unit Testing Components
- Mocking API calls
- Snapshot Testing
- End-to-End (E2E) Testing (e.g. Cypress)

UI Libraries & Styling

- Tailwind CSS
- Styled Components
- Emotion
- Material UI (MUI)
- Chakra UI
- React Bootstrap
- CSS Modules

Project Setup & Build

- Create React App (CRA)
- Vite + React
- ESLint + Prettier in React
- Folder and File Structure for Scaling
- Environment Variables (.env)
- Deploying React App (Vercel, Netlify, Firebase)
- Progressive Web Apps (PWAs)
- Server-Side Rendering (Next.js)
- Static Site Generation (Next.js)
- TypeScript in React